Chatgpt를 이용하여 만든 새로운 digital logic circuit 연습문제

ChatGPT

Plus 사용하기

1. 다음 상태표를 각각에 대하여, 상태도와 가능한 곳까지 타이밍 추적을 완성하라(입력 값이 알려지지 않은 후에도).

a.

q_1q_2	q_1q_2 (x=0)	q_1q_2 (x=1)	z (x=0)	z (x=1)
00	10	01	0	1
01	11	10	1	0
10	00	11	0	1
11	01	00	1	0

x : 1 0 1 1 0 1 0 0 1

q_1 : 1

q_2 : 0

z :

ChatGPT

Plus 사용하기

2. 다음 회로의 출력 파형을 timing diagram으로 완성하라.

a.

회로 조건:

- 두 개의 JK Flip-Flop 사용
- FF1의 입력: J = x, K = q_2
- FF2의 입력: J = q_1, K = x'
- 출력 z = q_1 XOR q_2
- 두 Flip-Flop은 같은 clock 사용
- 초기 상태: q_1q_2 = 00

입력 파형:

x : 0 1 1 0 1 0 0 1 1 0

clock : ↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑

q_1 :

q_2 :

z :

3. 다음 Flip-Flop 상태표로부터 Boolean Algebra를 이용하여 입력식을 도출하라.

a. JK Flip-Flop

현재상태 q_1q_2	다음상태 q_1q_2
00	01
01	11
10	00
11	10

요구사항:

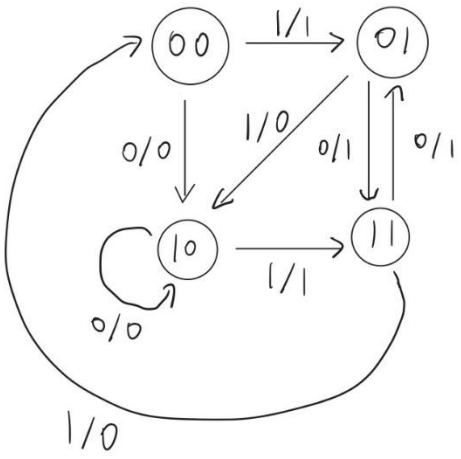
1. q_1에 대한 J_1, K_1 식 도출
2. q_2에 대한 J_2, K_2 식 도출
3. 최소화된 Boolean 식 제시

문제 풀이

문제 1

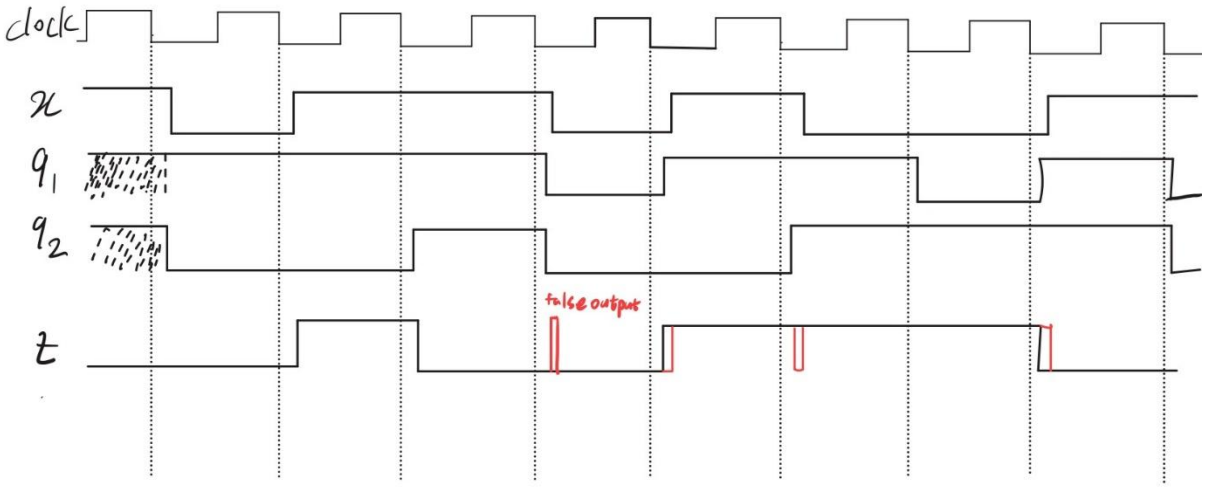
$q_1 q_2$	$q_1^* q_2^*$ (X=0)	$q_1^* q_2^*$ (x=1)	z (x=0)	z (x=1)
00	10	01	0	1
01	11	10	1	0
10	10	11	0	1
11	01	00	1	0

① State diagram



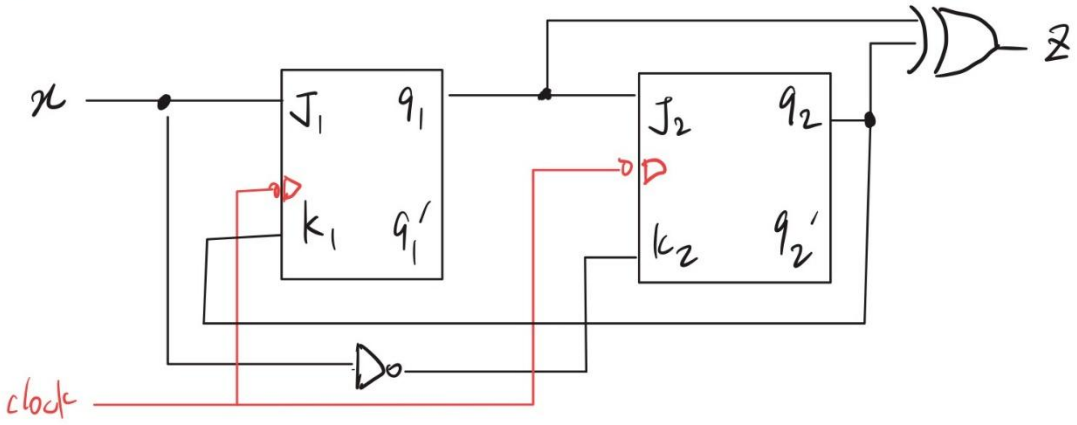
x	q_1	q_2	q_1^*	q_2^*	z
0	0	0	1	0	0
0	0	1	1	1	1
0	1	0	1	0	0
0	1	1	0	1	1
1	0	0	0	1	1
1	0	1	1	0	0
1	1	0	1	1	1
1	1	1	0	0	0

② timing trace



x	1	0	1	1	0	1	0	0	1
q_1	?	1	1	1	0	1	1	0	1	0
q_2	?	0	0	1	0	0	1	1	1	0
z	0	0	1	0	0	1	1	1	0

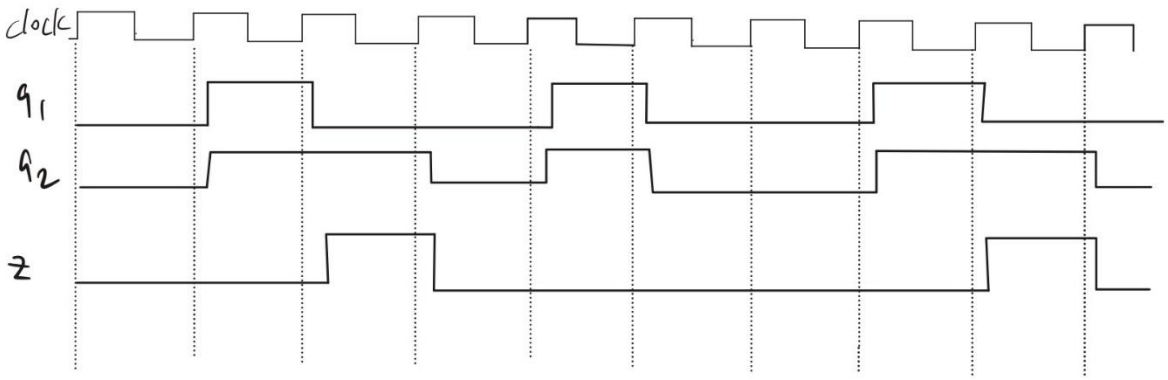
문제 2



x	q_1	q_2	q_1^*	q_2^*	Z
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	1	1	0
0	1	1	0	0	0
1	0	0	1	1	0
1	0	1	1	1	0
1	1	0	1	1	0
1	1	1	0	1	0

⇒

x	0	1	1	0	1	0	0	1	1	0
q_1	0	1	0	0	1	0	0	1	0	0
q_2	0	1	1	0	1	0	0	1	1	0
Z	0	0	1	0	0	0	0	0	1	0



문제 3

q_1	q_2	q_1^*	q_2^*
0	0	0	1
0	1	1	1
1	0	0	0
1	1	1	0

$\Rightarrow$

q_1	q_1^*	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

q_2	q_2^*	J	K
0	1	1	X
1	1	X	0
0	0	0	X
1	0	X	1

즉 ① $q_2=0 \rightarrow J_1=0$
$q_2=1 \rightarrow J_1=1$ $\Rightarrow$ $J_1=q_2$

② $q_2=0 \rightarrow K_1=1$
$q_2=1 \rightarrow K_1=0$ $\Rightarrow$ $K_1=q_2'$

③ $q_1=0 \rightarrow J_2=1$
$q_2=1 \rightarrow J_2=0$ $\Rightarrow$ $J_2=q_1'$

④ $q_1=0 \rightarrow K_2=0$
$q_1=1 \rightarrow K_2=1$ $\Rightarrow$ $K_2=q_1$

답: $J_1=q_2$ $K_1=q_2'$ $J_2=q_1'$ $K_2=q_1$